

Light

Newton corpuscles $\Rightarrow$ th
Particle Wave

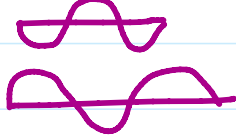
Reflection
Refraction
Dispersion
Interference
Diffraction
Polarization

2. Huygen Wave theory

wave

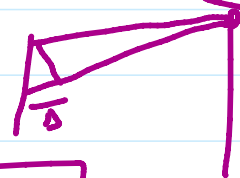
$\lambda, \nu, A, T, \Delta, \phi$ $\nu = \frac{c}{\lambda}$

$I \propto A^2$



$$\phi = \frac{2\pi}{\lambda} \times \Delta$$

m, p, v, s

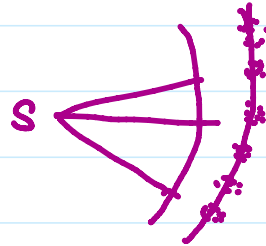


Longitudinal



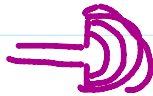
Transverse

ether x



Wave front

Secondary wavelet



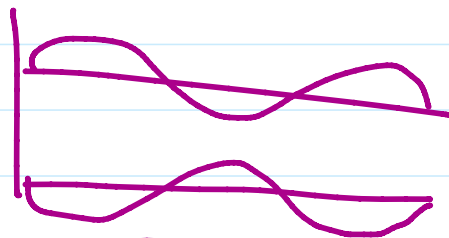
1. Plane

2. Cylindrical $\rightarrow$ {||}

3. Spherical $\rightarrow$ Pt. son



$\checkmark C1$



$D1 \checkmark$

Young $\gamma D S E$

$$\psi = \psi_1 + \psi_2 \quad - (1)$$

$$\psi_1 = a_1 \sin \omega t \quad - (2)$$

$$\psi_2 = a_2 \sin(\omega t + \phi) \quad - (3)$$

$$\psi = a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi$$

$$\psi = (a_1 + a_2 \cos \phi) \sin \omega t + (a_2 \sin \phi) \cos \omega t$$

$$a_1 + a_2 \cos \phi = A \cos \theta \quad - (4)$$

$$a_2 \sin \phi = A \sin \theta \quad - (5)$$

$$\psi = A \cos \theta \sin \omega t + A \sin \theta \cos \omega t$$

$$\psi = A \sin(\omega t + \theta) \quad - (6)$$

$$\psi = A \cos \theta \sin \omega t + A \sin \theta \cos \omega t$$

$$\psi = A \sin(\omega t + \theta) \quad - (6)$$

So add eq (6) & (5)

$$A^2 = a_1^2 + a_2^2 \cos^2 \phi + 2a_1 a_2 \cos \phi + a_2^2 \sin^2 \phi$$

$$A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi \quad - (7)$$

$$I \propto A^2$$

$$I = K A^2$$

$$I = K (a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi) \quad - (8)$$

C₂

$$\cos \phi = +1$$

$$\cos \phi = -1$$

$$I_{\max} = (a_1 + a_2)^2 \quad a_1 = a_2 = a$$

$$I_{\min} = (a - a)^2$$

$$I_{\min} = 0$$

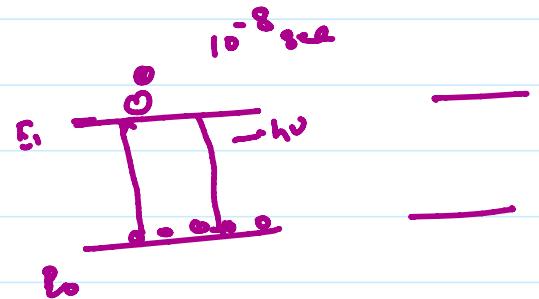
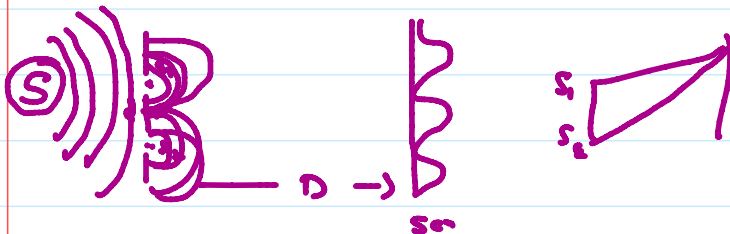
$$\phi = 2n\pi \quad n = 0, 1, \dots$$

$$\phi = (2n+1)\pi \quad n = 0, 1, 2$$

$$\Delta = n\lambda \quad I_{\max} = 4a^2$$

$$\Delta = (2n+1) \frac{\lambda}{2} \quad - (10)$$

= - (9)



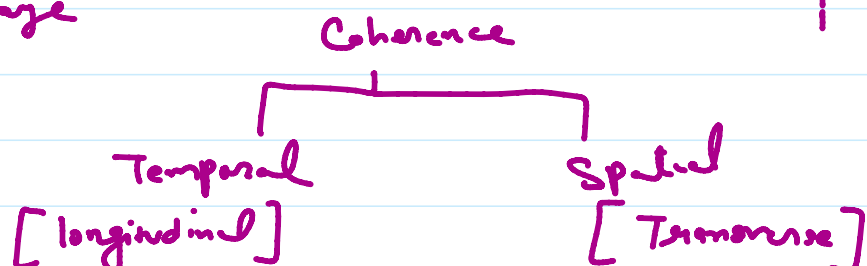
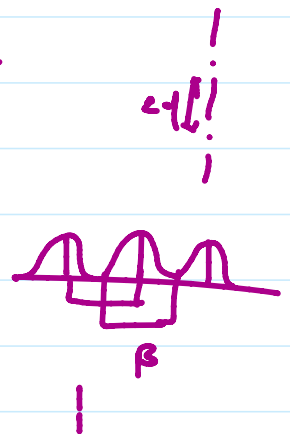
Essential Interference

$$\beta = \frac{\lambda D}{2d} \quad \checkmark$$

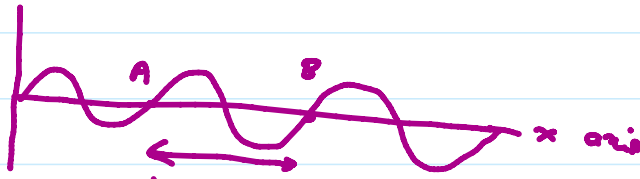
1. Coherent
2. Wavelength / freq
3. Amplitude equal
4. Same direction
5. Monochromatic
6. λ should be large
7. D should be large

8. $2d$ must be small

9. Background VIBRATOR dark



[longitudinal]



$$t' - t = \Delta L$$

Cohherence time

$\Delta L = \text{Cohherence length}$

$$\Delta L = c \Delta t$$

Classes of Interference

1. Division of wavefront $\rightarrow$ YDSE, Fresnel Biprism
2. Division of Amplitude $\rightarrow$ partial reflection/transmission

Thin film Interference

R.P

$$a_1: a_1^2 \lambda: a_1^4 \lambda^2: \dots$$

$$1: t^2: t^2 \lambda^2: \dots$$

T.P

$$a_1^2: a_1^2 \lambda^2: a_1^2 \lambda^3: \dots$$

$$1: \lambda^2: \lambda^3: \dots$$

$$\mu = 1.5$$

$$r_1 = \frac{\mu - 1}{\mu + 1}$$

$$r_1 = \frac{1.5 - 1}{1.5 + 1} = 0.2$$

$$r_1^2 = 0.04 < 1$$

$$t^2 = 1 - r_1^2$$

$$\Rightarrow t^2 = 0.96 \approx 1$$

1.2. Stokes Law

$$\Delta = 2\mu t \cos \alpha$$

(R.P)

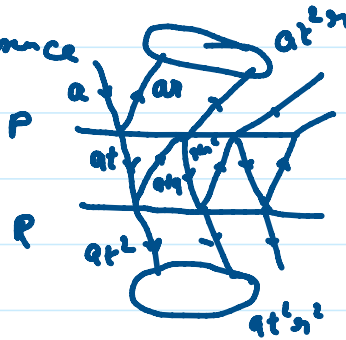
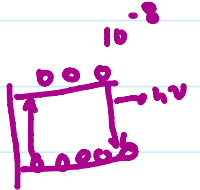
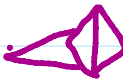
$$\Delta = 2\mu t \cos \alpha + \frac{\lambda}{2}$$

$$\Delta = n\lambda$$

T.P

$$\Delta = 2\mu t \cos \alpha$$

$$\Delta = n\lambda$$



Reflected Part

Transmitted Part



π

$$2\mu t \cos \theta + \frac{\lambda}{2} = n\lambda$$

$$2\mu t \cos \theta = (2n-1) \frac{\lambda}{2} \quad \text{--- (1)}$$

$$DZ: \Delta = 2\mu t \cos \theta + \frac{\lambda}{2}$$

$$\Delta = (2n+1) \frac{\lambda}{2}$$

)))

$$2\mu t \cos \theta + \frac{\lambda}{2} = (2n+1) \frac{\lambda}{2}$$

$$2\mu t \cos \theta = n\lambda \quad \text{--- (2)}$$

Newman's Ring



$$\Delta = 2\mu t \cos \theta$$

$$n=1$$

$$\cos \theta = 1$$

$$\Delta = 2t$$

$$CA = R-t$$

$$AO = t$$

$$CP = R$$

$$CP^2 = CA^2 + AP^2$$

$$R^2 = (R-t)^2 + x^2$$

$$R^2 = R^2 + t^2 - 2Rt + x^2$$

$$x^2 = 2Rt$$

$$x_n^2 = 2Rt$$

$$2t = n\lambda$$

$$x_n^2 = Rn\lambda$$

$$D_n^2 = 4Rn\lambda$$

$$n^2 = 4R(n+p)\lambda$$

Let $t = n^{\text{th}}$ dark

$$2t = n\lambda$$

$$2\mu t \cos \theta = n\lambda \quad \checkmark \quad \text{--- (3)}$$

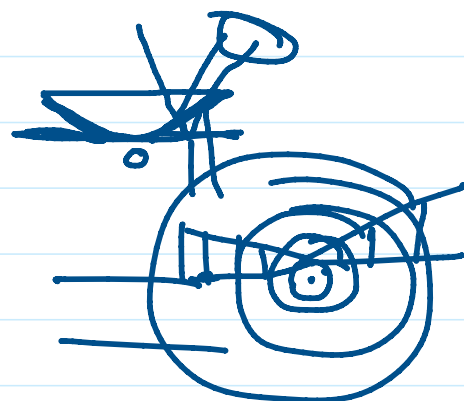
$$\Delta = 2\mu t \cos \theta$$

$$\Delta = (2n+1) \frac{\lambda}{2}$$

$$2\mu t \cos \theta = (2n+1) \frac{\lambda}{2} \quad \text{--- (4)}$$



wedge shaped



$$CZ$$

$$2\mu t \cos \theta + \frac{\lambda}{2} = n\lambda$$

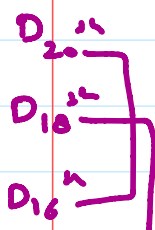
$$2t \cos \theta + \frac{\lambda}{2}$$

$$2t + \frac{\lambda}{2} = n\lambda$$

$$2t = (2n-1) \frac{\lambda}{2}$$

DZ

$$2t = n\lambda$$



$$D_n^2 = 4Rn\lambda$$

$$D_{n+p}^2 = 4R(n+p)\lambda$$

$$D_{n+p}^2 - D_n^2 = 4Rp\lambda$$

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \quad \checkmark \checkmark$$

2. To find Refractive Index of the given liquid

We know that

$$D_n^2 = \frac{4Rn\lambda}{\mu} \quad [\text{for any liquid}]$$

$$\& \quad D_n^2 = 4Rn\lambda \quad [\text{for air film}]$$

Equating both we get

$$\mu D_n^2(\text{liq}) = D_n^2(\text{air})$$

$$\therefore \mu = \frac{D_n^2(\text{air})}{D_n^2(\text{liq})} \quad \left. \vphantom{\frac{D_n^2(\text{air})}{D_n^2(\text{liq})}} \right\} \text{ formula to find refractive index}$$

Difference between Interference and Diffraction:

Interference	Diffraction
1. Interference is due to the superposition of two different wave trains coming from coherent sources.	1. Diffraction is due to the superposition of secondary wavelets from the different parts of the same wavefront.
2. Fringe width is generally constant.	2. Fringes are of varying width.
3. All the maxima have the same intensity.	3. The maxima are of varying intensities.
4. There is a good contrast between the maxima and minima.	4. There is a poor contrast between the maxima and minima.

Classes of Diffraction

Fraunhofer diffraction	Fresnel diffraction
Source and the screen are far away from each other.	Source and screen are not far away from each other.
Incident wave fronts on the diffracting obstacle are plane.	Incident wave fronts are spherical.
Diffracting obstacle give rise to wave fronts which are also plane.	Wave fronts leaving the obstacles are also spherical.
Plane diffracting wave fronts are converged by means of a convex lens to produce diffraction pattern.	No Convex lens is needed to converge the spherical wave fronts.